MILITARY SPECIFICATION

CALIBRATOR, MAGNETIC; TS-85(*)/ASQ

This specification has been approved
by the Bureau of Aeronautics, Department of the Navy

## 1. SCOPE AND CLASSIFICATION

1.1 Scope - The equipment covered by this specification shall be a magnetic calibrator to be used as a standard reference to measure the magnetic field strength at various distances from its magnetic center in calibrating the AN/ASQ-8 equipment.

1.2 Classification - The equipment shall be of one type and shall consist of the following:

| Quan Per Equipment | Nomenclature | Type Designation | Max Allow. Size (in.) W H L/D | Max Allow. Wt Per Unit (lb) | Applicable Paragraph |
|---|---|---|---|---|---|
| 1 | Magnetic Calibrator | TS-85(*)/ASQ | 8 10 8 | 15 | 3.3.1 |
| 1 | Power Cord | CX-3277/U | 72 | | 3.3.1.6 |

* The parenthesis in the type designation shall be replaced by a letter furnished by the procuring agency upon application by the contractor in accordance with the requirements of paragraph 3.2.9 after each contract has been awarded. The type number with the furnished letter shall be used on nameplates, shipping records, and instruction books, as applicable.

## 2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, shall form a part of this specification to the extent specified herein:

2.2 Specifications:

Military

| | |
|---|---|
| MIL-T-945 | Test Equipment, For Use With Electronic Equipment; General Specification |
| JAN-P-658 | Packaging and Packing of Electrical Equipment and Spare Parts (Electronic, Electrical, and Electromechanical) |
| MIL-N-18307 | Nameplates and Nomenclature; Airborne Electronic Equipment |

2.3 Standards

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |

2.4 Availability of Applicable Documents:

(1) When requesting applicable documents, refer to both title and number. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable specifications and standards may be obtained upon application to the Commanding Officer, Naval Air Station, Johnsville, Pennsylvania, Attn: Technical Records Division.



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2.5 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring agency.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - The equipment shall fulfill the following requirements for parts and materials:

3.1.1 Design Objectives - Minimum size and weight, simplicity of operation, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives in the design of the equipment specified herein. The use of materials, parts, and processes other than those required by Specification MIL-T-945 shall be investigated, and when it appears that a substantial reduction in size and weight or improvement in simplicity of design, performance, or reliability can be realized by their substitution, a request for approval of a nonstandard part shall be submitted to the procuring agency for consideration. Each request shall be accompanied by complete supporting data.

3.1.2 Selection of Parts and Materials - The fulfillment of major design objectives shall govern the selection of parts and materials. All parts and materials shall meet the requirements of Specification MIL-T-945; however, if the contractor can demonstrate that the use of such parts and materials will not fulfill the major design objectives, then parts and materials shall be used which most nearly meet or exceed the requirements of Specification MIL-T-945.

3.1.3 Approval for Nonstandard Parts and Materials - Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-T-945. Requests for approval must be forwarded to procuring agency by the contractor before parts are purchased or fabricated.

3.1.4 Workmanship - Workmanship, fabrication, and manufacturing process shall conform to the requirements of Specification MIL-T-945.

3.2 General Design Requirements

3.2.1 General - This equipment shall fulfill all applicable design and performance requirements of Specification MIL-T-945 unless otherwise specified.

3.2.2 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions, or reasonable combinations of service conditions, as specified in MIL-T-945 except that the requirements specified shall apply only within the temperature range from -40° to +55° C.

3.2.3 Size and Weight - The over-all dimensions of the equipment shall not be greater than 8 inches wide, 8 inches deep, and 10 inches high. The total weight shall not be greater than 15 pounds.

3.2.4 Operating Power - The equipment shall operate as specified herein from 28 V d-c ± 1.5 V.

3.2.5 Operating Life

3.2.5.1 Reliable Operating Life - The equipment shall be capable of operation, as specified herein, for a period of 200 hours without requiring servicing or replacement of parts. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.



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3.2.5.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing or replacement of parts. The life of such parts and parts requiring replacement within this interval shall be specified by the manufacturer.

3.2.6 Operational Stability - The equipment shall operate with optimum performance for at least 200 hours, continuously or intermittently, without necessity for adjustment of controls other than those located on the front panel.

3.2.7 Cables and Connectors - Cables and connectors shall be in accordance with the requirements of Specification MIL-T-945.

3.2.8 Interchangeability - Mechanical and electrical interchangeability shall exist between similar assemblies, subassemblies, and replaceable parts regardless of the manufacturer or supplier. Interchangeability for the purposes of this paragraph does not mean identity, but requires that a substitution of such like assemblies, subassemblies, and replaceable parts be easily effected without physical or electrical modifications to any part of the equipment including cabling, wiring, and mounting. In the design of the equipment, provisions shall be made for design tolerance to accommodate various articles such as tubes, resistors, and other components having the limiting dimensions and characteristics set forth in the specifications for the particular component involved without departure from the specified performance.

3.2.9 Nameplates and Nomenclature - Nameplates and nomenclature shall be in accordance with the requirements of Specification MIL-N-18307.

3.3 Detail Design Requirements

3.3.1 Magnetic Calibrator TS-85A/ASQ - The magnetic calibrator shall consist of a magnet mounted within a nonmagnetic housing with all necessary operating controls mounted on the front panel. The unit shall be capable of being withdrawn from the cabinet without disconnecting any leads. The unit as withdrawn from the cabinet shall be capable of being placed on a bench for servicing without danger of damage by bending or breaking of parts, etc. It shall be suitable for use in calibration as a standard reference to measure the magnetic field strength at various distances from its magnetic center. It is not intended for use in flight.

3.3.1.1 Calibration Magnet - The magnet shall be a rotatable round permanent magnet and have a magnetic moment strength of 1000 C.G.S. units. This moment strength shall be maintained to within ±2 percent. The north seeking pole of the magnet shall be marked "N".

3.3.1.1.1 Form Factor - The over-all dimensions of the calibration magnet shall not be greater than 0.25 inch in diameter, and 3 inches long.

3.3.1.2 Magnet Cover - The magnet shall be provided with a suitable nonmagnetic protective cover. This cover shall completely enclose the magnet but shall leave it accessible for removal by screwdriver.

3.3.1.3 Rotating Speed - The magnet shall be rotated at speeds of 0.01, 0.05, 0.1, 0.25, 0.5, 1.0 and 2.0 cps. Each of these speeds shall be accurate to within ±10 percent. Suitable provisions shall be made on the top panel for changing speeds in increments, and a graduated scale in cycles per second shall be furnished on the same panel, to facilitate correct reading. The scale characters shall be permanently marked. The ±10 percent accuracy required must be realized when the operating voltage is varied within allowable limits.

3.3.1.4 Drive - A gear train similar to the drive used in the signal generator on the AN/ASM-3 (currently available from ASO) may be used.

3.3.1.5 Power - The magnetic calibrator shall be powered with a d-c motor operating on 28 V d-c. The motor shall be of such constant speed that the over-all speed tolerance noted in paragraph 3.3.1.3 will be maintained.

3.3.1.6 Power Cord CX-3277/U - The power cord shall consist of a 6-foot length of 2-conductor cable terminated at one end with a connector capable of mating with a type AN3102-10SL-3P connector. The other end shall be terminated with a 2 prong polarized plug.



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3.3.1.7 Measuring Tape - A corrosion resistant, nonmagnetic durable metal tape of the self retracting type shall be provided, and shall be calibrated from the same center as the rotating magnet. The tape shall be a minimum 3/8-inch and a maximum 3/4-inch wide. It shall be graduated in feet and inches on one edge with 1/8-inch increments, and in gammas on the other edge. Both scales shall be on the same side of the tape. The minimum reading shall be 0.5 gamma and the maximum reading shall be 50 gammas with the following gamma scale subdivisions marked: 0.4, 0.6, 0.8, 1.0, 2.0, 4.0, 6.0, 8.0, 10, 20, 30, 40, 50. All gamma scales shall be based on amplitude or peak signals with the magnet and magnetometer in the unfavorable position. This information, along with simple sketch, shall be prominently displayed on the front panel. The 1 gamma marking shall be located at the extreme "pulled out" position of the tape and shall correspond with the maximum feet and inches on the other edge. The end of the tape shall be accessible from the top of the panel, and it shall be able to be pulled out to its maximum length using only finger pressure. All characters, lines, etc., shall be desirable to read and shall be permanently marked.

3.3.1.8 Transit Case - The transit case shall be a combination type in accordance with Specification MIL-T-945. The case shall contain an AN 3106-10SL-3P receptacle to mate with the connector on the power cord.

3.3.1.9 Rubber Feet - The unit shall be provided with four rubber feet on the bottom of the case, and on other surfaces as required, so that the rotating magnet will be rotatable in 90-degree increments in any plane.

3.3.1.10 Nonmagnetic Materials - All materials used herein shall be nonmagnetic, except the magnet and the motor.

3.3.1.11 Fuses - The equipment shall be provided with a fuse and fuse holder in accordance with Specification MIL-T-945.

## 4 METHODS OF SAMPLING, INSPECTION, AND TEST

4.1 Classification - The equipment specified herein shall be subject to preproduction and inspection tests as follows:

(1) Preproduction Tests - Preproduction tests shall be made on a sample(s) representative of the production equipments, to determine that the production equipments will meet the requirements of this specification and the applicable subsidiary specifications.

(2) Inspection Tests - Inspection tests shall be made on production test sets submitted for acceptance under the contract to determine conformance with the requirements of this specification and the applicable subsidiary specifications and to determine that the test sets are equivalent in performance and construction to the approved preproduction sample. Any production prior to approval of the preproduction sample shall be at the contractor's risk.

4.2 Preproduction Tests - Preproduction tests shall be conducted on two preproduction models representative of the production equipment as follows:

4.2.1 Scope of Preproduction Tests - Preproduction tests shall be conducted at a laboratory designated by the procuring agency, and shall be in accordance with paragraph 4.2.1.2 of Specification MIL-T-945. The procuring agency, however, reserves the right to make additional preproduction tests other than those outlined in Specification MIL-T-945 to insure compliance with this specification and the contract.

4.2.2 Presubmission Tests - Prior to the submission of any models for preproduction testing, the contractor shall furnish facilities for and shall conduct tests as required in paragraph 4.2.4 of Specification MIL-T-945. In addition, the data required in paragraph 4.2.3 of Specification MIL-T-945 shall be submitted to the procuring agency for review and approval prior to submission of any models.

4.2.3 Approval of Preproduction Models - Approval of preproduction model shall be by the procuring agency upon satisfactory completion of all tests. No additional models shall be delivered prior to the approval of the preproduction model. Prefabrication of any equipments prior to the approval of the preproduction model is at the contractor's own risk. One approved preproduction model will be returned to the contractor for his use in the fabrication and testing of the models to be submitted for acceptance under the contract. The preproduction model shall not be considered as one of the models under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production model.



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4.3 Inspection Tests - The contractor shall furnish all models and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the cognizant Naval Inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the procuring agency. The contractor shall furnish test reports in duplicate, showing quantitative results for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.3.1 Scope of Inspection Tests - Inspection tests shall meet all the requirements of MIL-T-945, and shall be of the following two types:

(1) Individual Tests - Individual tests shall be conducted on each model submitted for acceptance under the contract.

(2) Sampling Tests - Sampling tests shall be conducted on the quantity indicated in paragraph 4.3.1.1. Models submitted for sampling tests shall be selected by the Government Inspector and shall first have passed the individual tests. Sampling tests shall be more extensive than the individual tests and may include any of the tests listed under preproduction tests which are deemed necessary by the inspector to determine that production models continue to be equivalent in performance and construction of the approved preproduction models.

4.3.1.1 Sampling Test Schedule - Sampling Tests shall be performed on the following quantity of equipment unless otherwise stated in the contract or approved in writing by the procuring service.

| Quantity of Equipment | Quantity of Equipment to be tested |
|---|---|
| First 10 equipments | 1 |
| Next 10 to 100 | 2 |
| Next 100 to 200 | 1 |
| Over 200 | 1 for each additional 200 |

4.4 Procedures - The procedures for conducting presubmission testing of preproduction models and inspection testing of production models shall be determined by the contractor. These test procedures shall be subject to review and approval by the procuring agency prior to the initiation of any testing. Use Specification MIL-T-18303 as a guide in the preparation of Test Procedures.

4.5 Acceptance of Production Models - Acceptance of all production models shall be by the cognizant Naval Inspector.

## 5. PACKAGING AND PREPARATION FOR SHIPMENT

5.1 Packaging - The equipment and spare parts shall be preserved, packaged, and packed in accordance with the requirements of Specification JAN-P-658 and shall be marked in accordance with the requirements of Specification MIL-STD-129. Suitable interior packing, blocking, and bracing shall be used to prevent shifting in the exterior shipping container and shock from handling in transit.

## 6. NOTES

6.1 Ordering Information - Invitation for bids and contracts or orders shall specify the title, number, and date of this specification.

PATENT NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.



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